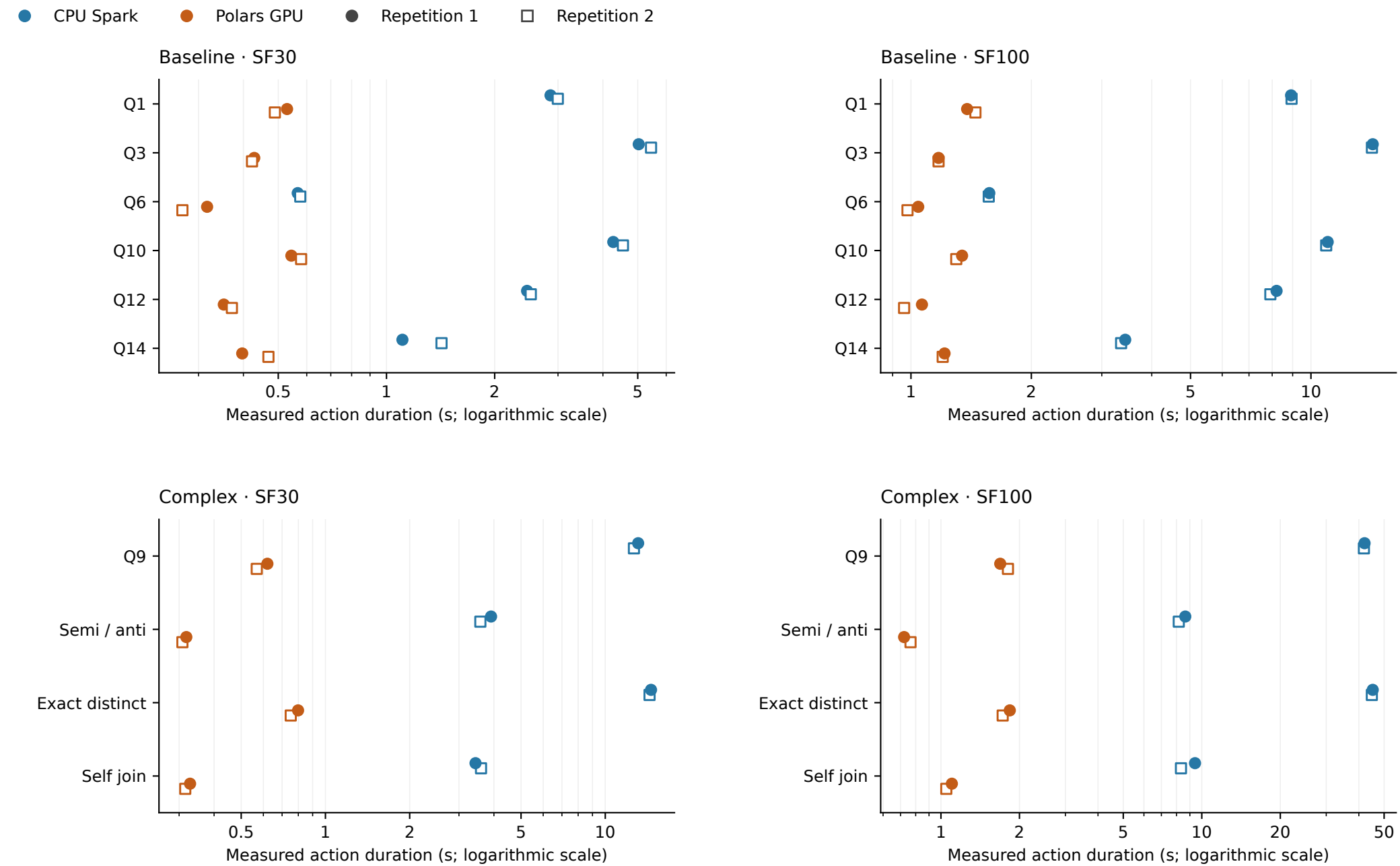
Expansion · pair B: CPU Spark + Polars GPU

Separate descriptive extension · COMPLETE



Each point is one measured action from a fresh concurrent wave; no averaging, confidence intervals or tests. Expansion has two repetitions per cell; pilot is shown separately. Common queries only. Pair B complex excludes Q13 and window: its CPU-only measurements remain unpaired diagnostics. Pairs have different competitors and do not rank the two GPU engines. Action timers exclude setup and rendezvous waits.